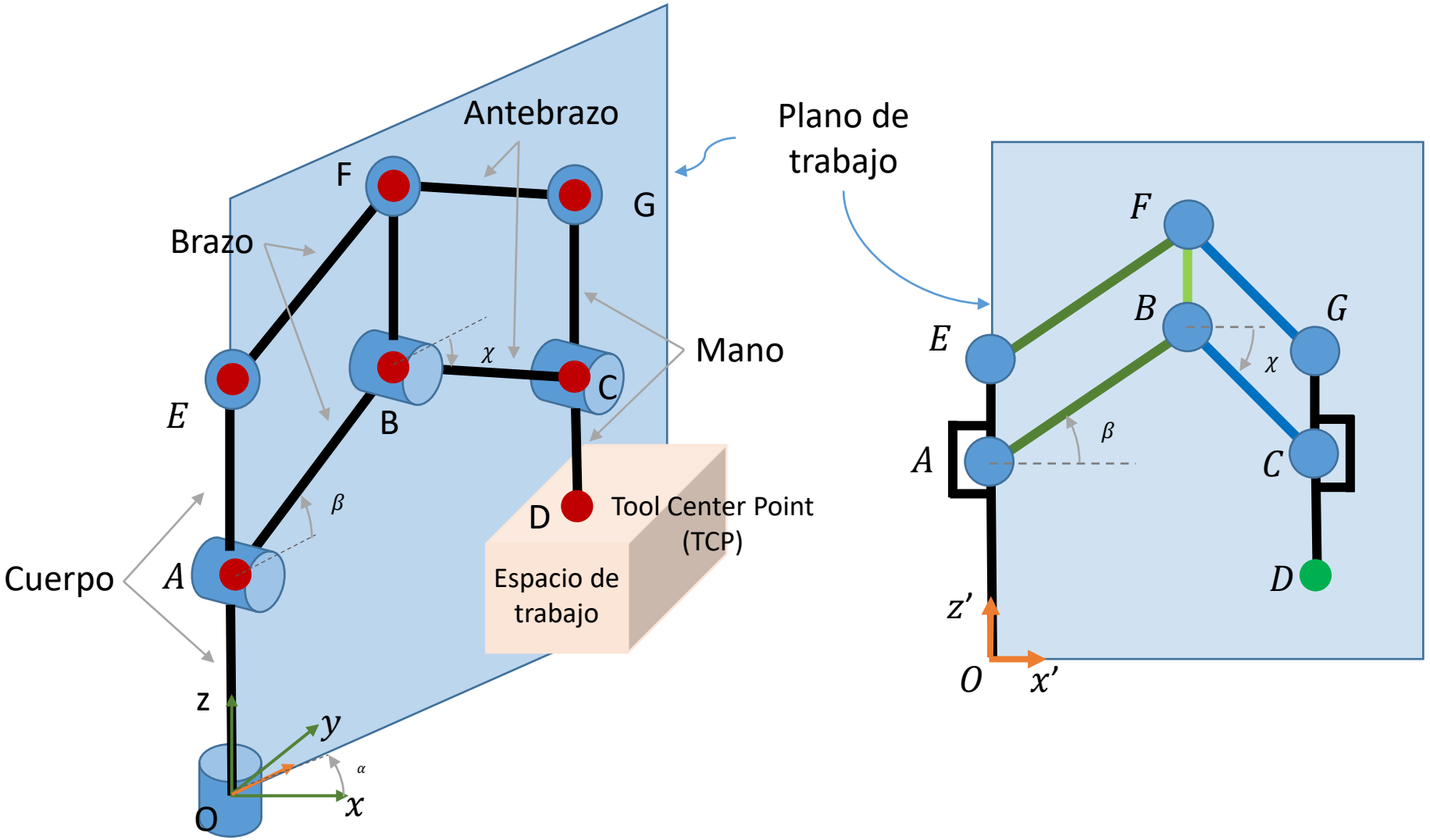
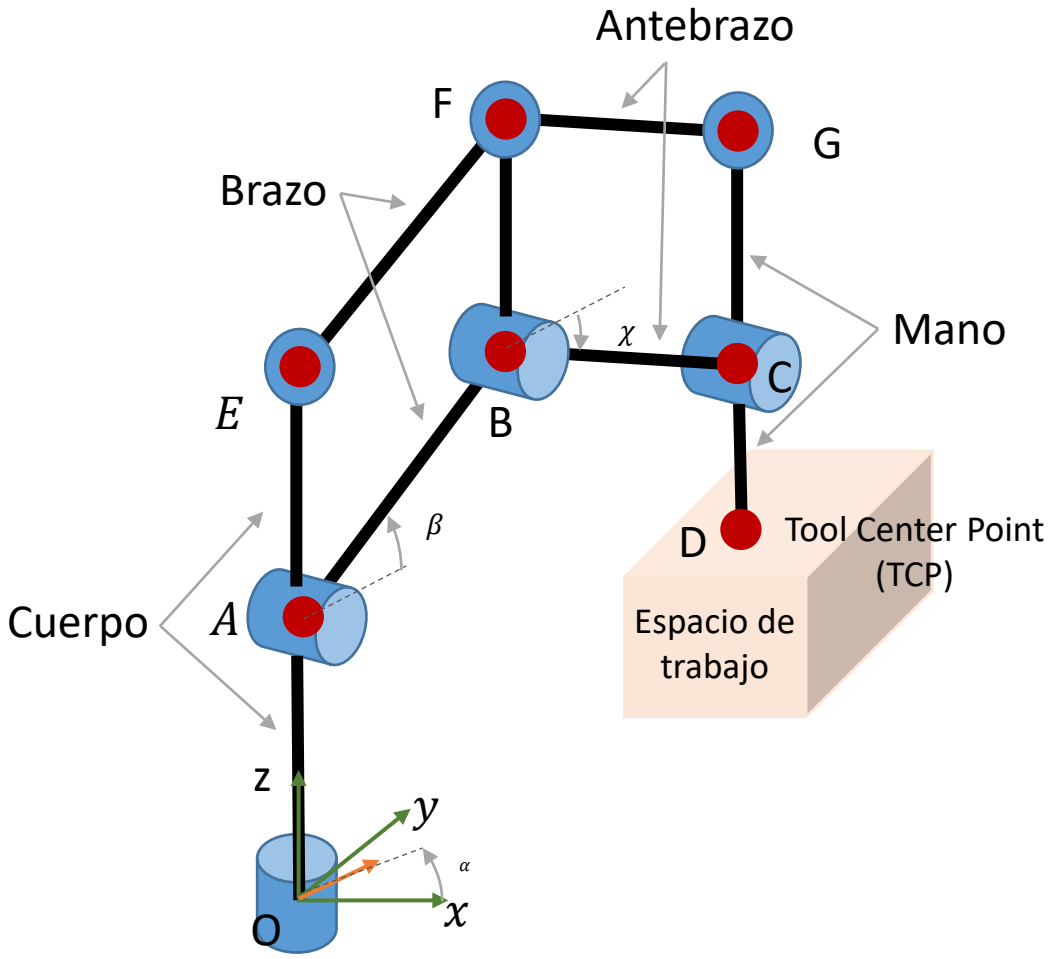
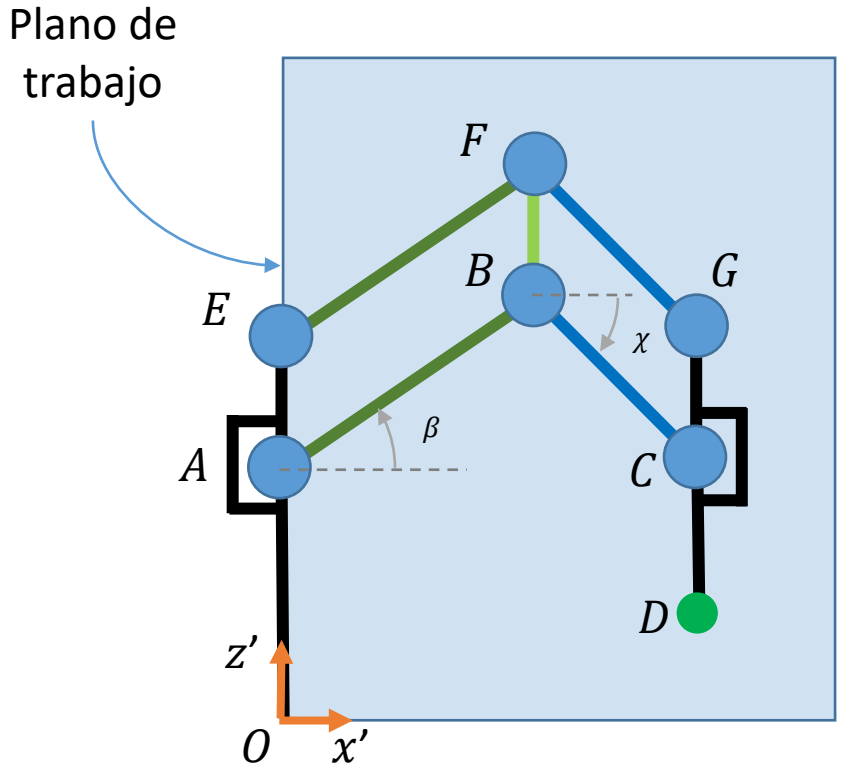
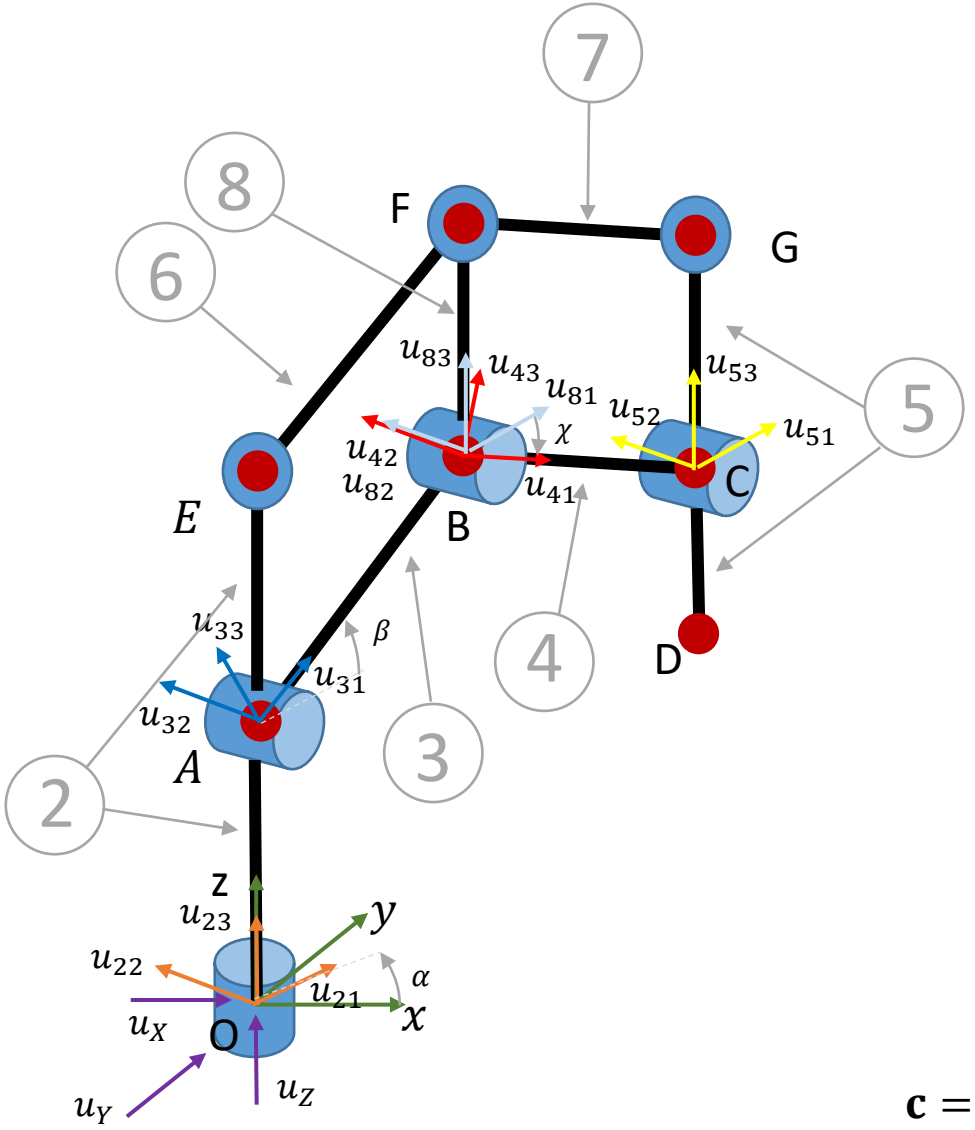

Antebrazo
Plano de trabajo
F
G
Brazo
Mano
E
B
C
χ
β
D
Tool Center Point (TCP)
Cuerpo
A
Espacio de trabajo
z
y
x
O
α
F
B
G
E
χ
β
A
C
D
z'
O
x'


Antebrazo
F
G
Brazo
Mano
χ
E
B
C
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Tool Center Point
(TCP)
Cuerpo
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Espacio de
trabajo
z
y
α
x
O


Plano de trabajo
F
B
G
E
χ
β
A
C
D
z'
O
x'



Vectores unitarios de las bases

Vectores de posición de los puntos básicos

Coordenadas relativas

(15*3)+(7*3)+3=69 coordenadas generalizadas

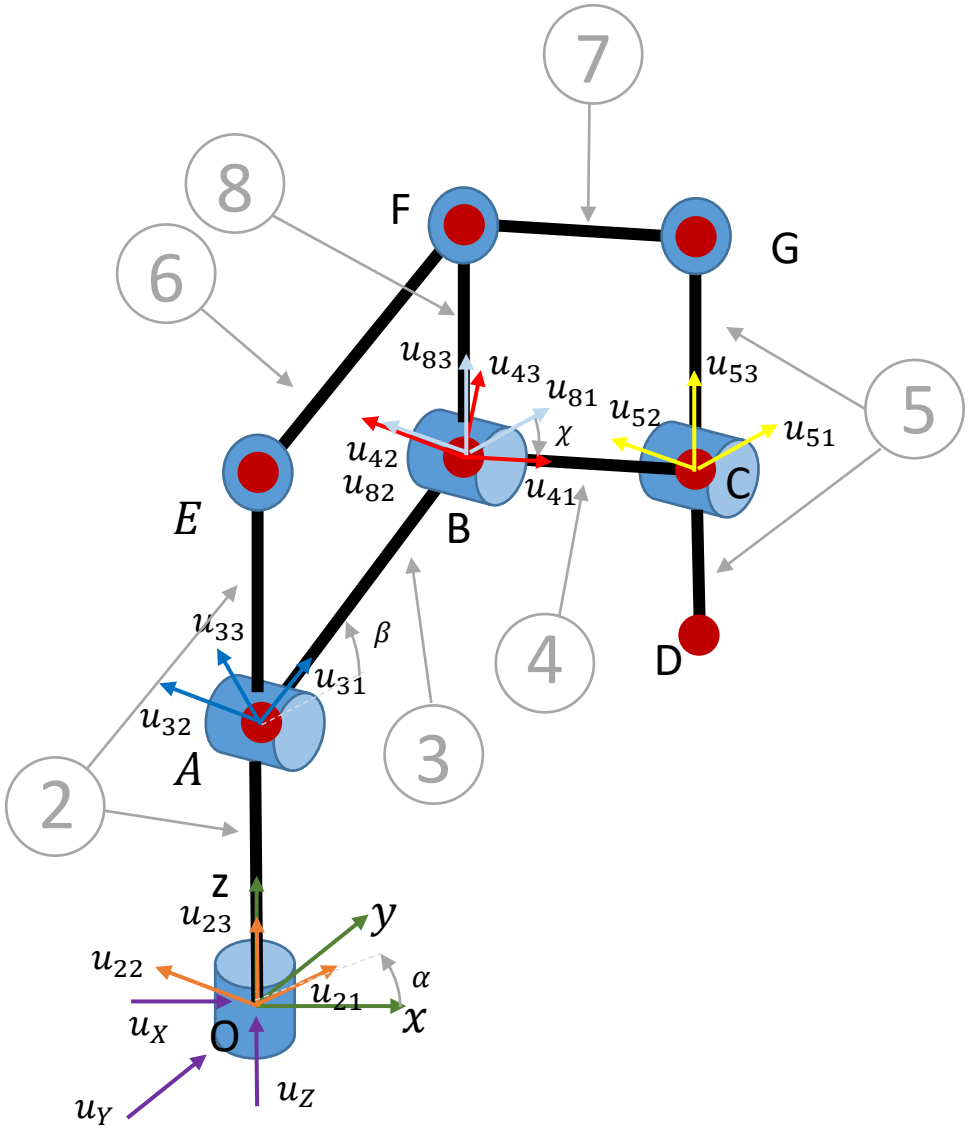
3 coordenadas independientes

66 ecuaciones de restricción + 8 redundantes

Sólido rígido 1 es el bastidor

$$\mathbf{c} = [\mathbf{c}_{A2}, \mathbf{c}_{E2}, \mathbf{c}_{B3}, \mathbf{c}_{C4}, \mathbf{c}_{D5}, \mathbf{c}_{G5}, \mathbf{c}_{F8}, L_{EF}, L_{FG}, \mathbf{r}_O, \mathbf{u}_X, \mathbf{u}_Y, \mathbf{u}_Z]$$

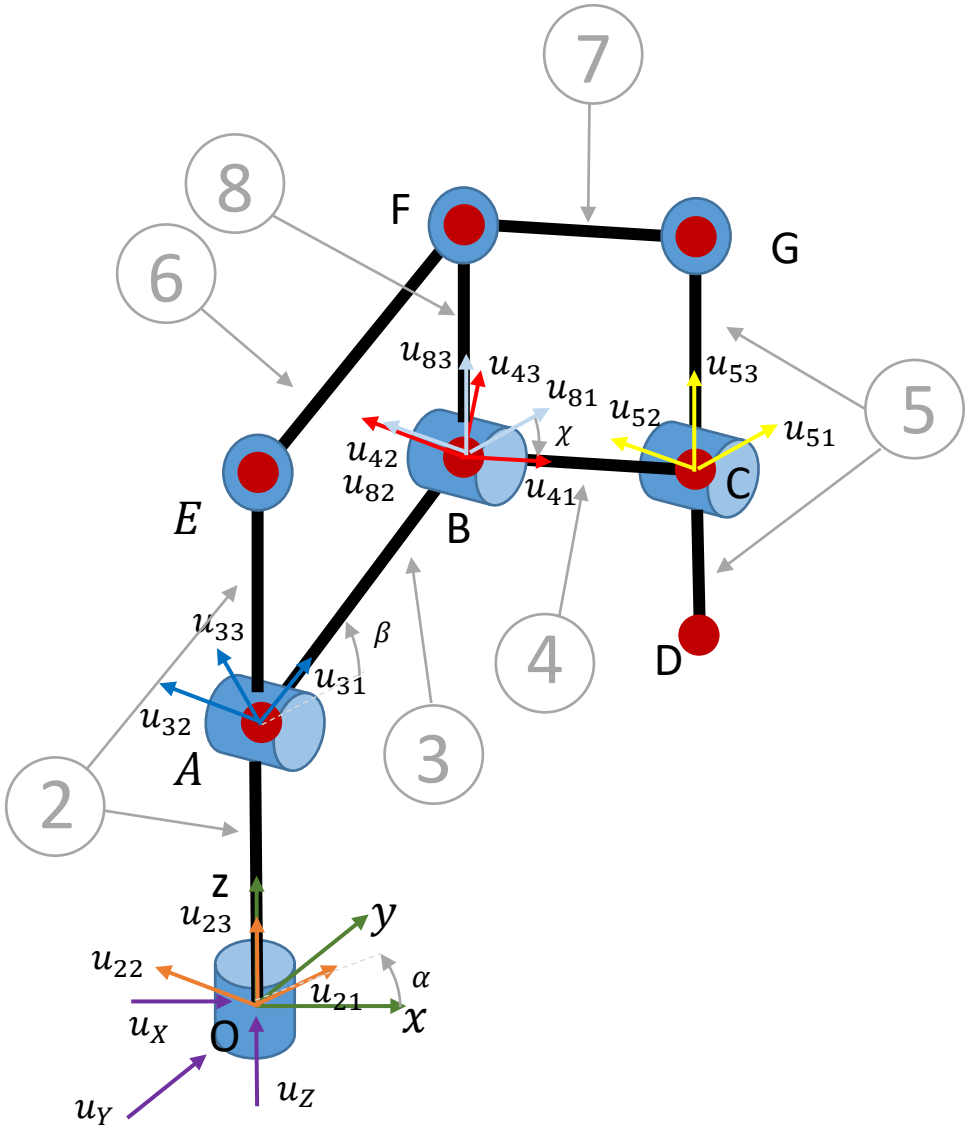
$$\mathbf{q} = [\mathbf{u}_{21}, \mathbf{u}_{22}, \mathbf{u}_{23}, \mathbf{r}_A, \mathbf{r}_E, \mathbf{u}_{31}, \mathbf{u}_{32}, \mathbf{u}_{33}, \mathbf{r}_B, \mathbf{u}_{41}, \mathbf{u}_{42}, \mathbf{u}_{43}, \mathbf{r}_C, \mathbf{u}_{51}, \mathbf{u}_{52}, \mathbf{u}_{53}, \mathbf{r}_D, \mathbf{r}_G, \mathbf{u}_{81}, \mathbf{u}_{82}, \mathbf{u}_{83}, \mathbf{r}_F, \alpha, \beta, \chi]^T$$



Constante	Puntero
$\mathbf{c}_{A2}$	1,2,3
$\mathbf{c}_{E2}$	4,5,6
$\mathbf{c}_{B3}$	7,8,9
$\mathbf{c}_{C4}$	10,11,12
$\mathbf{c}_{D5}$	13,14,15
$\mathbf{c}_{G5}$	16,17,18
$\mathbf{c}_{F8}$	19,20,21
L_{EF}	22
L_{FG}	23
$\mathbf{r}_O$	24,25,26
$\mathbf{u}_X$	27,28,29
$\mathbf{u}_Y$	30,31,32
$\mathbf{u}_Z$	33,34,35

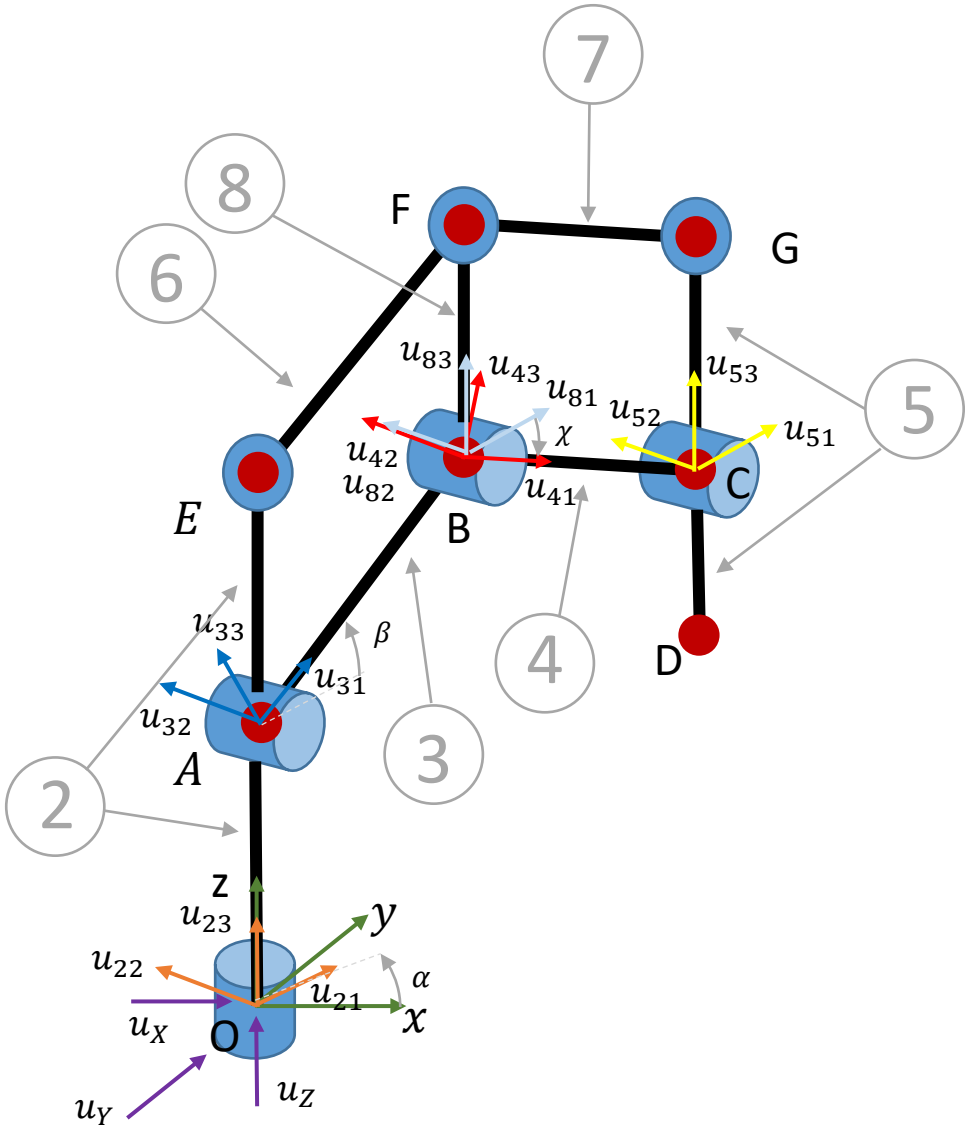
Coordenada	Puntero
$\mathbf{u}_{21}$	1,2,3
$\mathbf{u}_{22}$	4,5,6
$\mathbf{u}_{23}$	7,8,9
$\mathbf{r}_A$	10,11,12
$\mathbf{r}_E$	13,14,15
$\mathbf{u}_{31}$	16,17,18
$\mathbf{u}_{32}$	19,20,21
$\mathbf{u}_{33}$	22,23,24
$\mathbf{r}_B$	25,26,27
$\mathbf{u}_{41}$	28,29,30
$\mathbf{u}_{42}$	31,32,33
$\mathbf{u}_{43}$	34,35,36
$\mathbf{r}_C$	37,38,39
$\mathbf{u}_{51}$	40,41,42
$\mathbf{u}_{52}$	43,44,45
$\mathbf{u}_{53}$	46,47,48
$\mathbf{r}_D$	49,50,51

Coordenada	Puntero
$\mathbf{r}_G$	52,53,54
$\mathbf{u}_{81}$	55,56,57
$\mathbf{u}_{82}$	58,59,60
$\mathbf{u}_{83}$	61,62,63
$\mathbf{r}_F$	64,65,66
α, β, χ	67,68,69



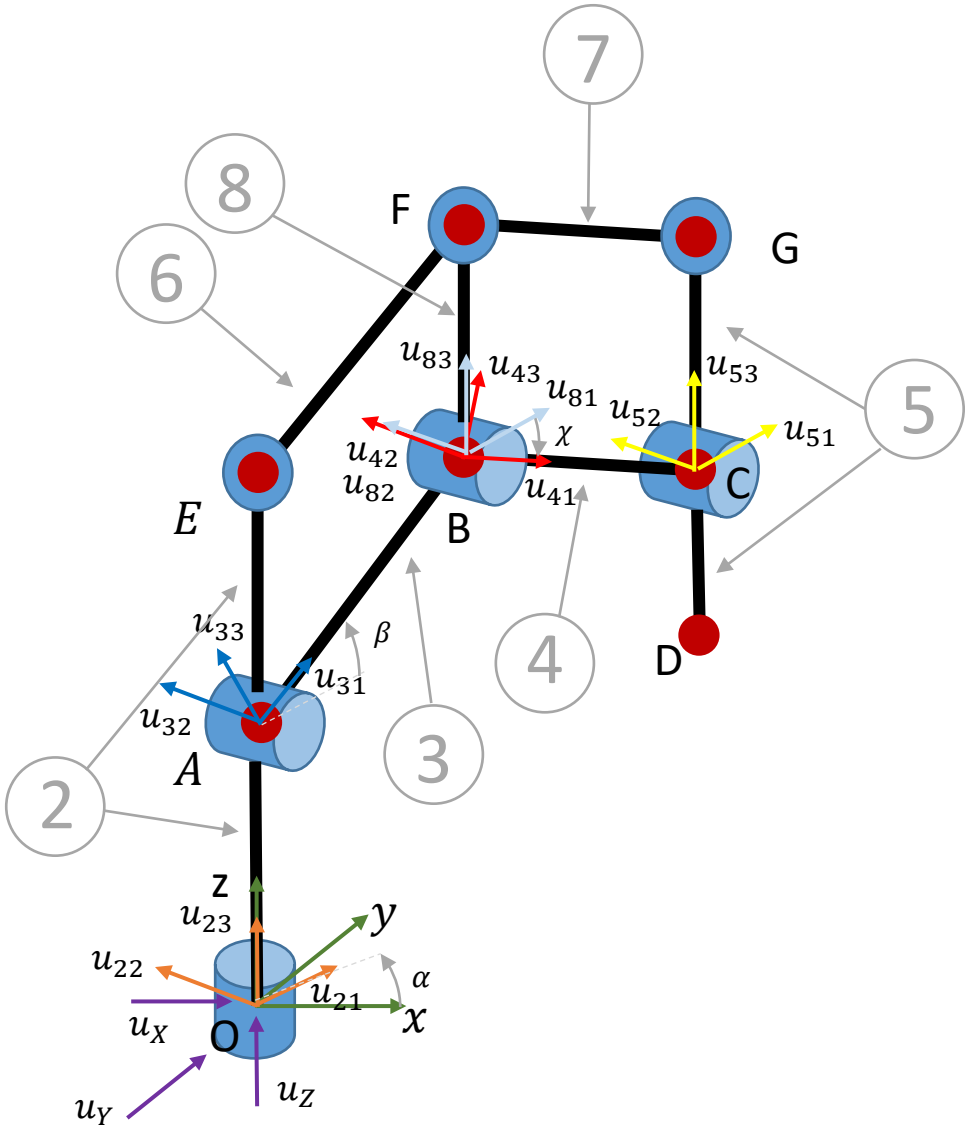
Relación geométrica	Ecuaciones	Número de ecuaciones
Sistema de referencia local de 2	$\lvert\mathbf{u}_{21}\rvert = 1$ $\lvert\mathbf{u}_{22}\rvert = 1$ $\lvert\mathbf{u}_{23}\rvert = 1$ $\mathbf{u}_{21} \cdot \mathbf{u}_{22} = 0$ $\mathbf{u}_{21} \cdot \mathbf{u}_{23} = 0$ $\mathbf{u}_{22} \cdot \mathbf{u}_{23} = 0$	6
Sistema de referencia local de 3	$\lvert\mathbf{u}_{31}\rvert = 1$ $\lvert\mathbf{u}_{32}\rvert = 1$ $\lvert\mathbf{u}_{33}\rvert = 1$ $\mathbf{u}_{31} \cdot \mathbf{u}_{32} = 0$ $\mathbf{u}_{31} \cdot \mathbf{u}_{33} = 0$ $\mathbf{u}_{32} \cdot \mathbf{u}_{33} = 0$	6

Total ecuaciones=12



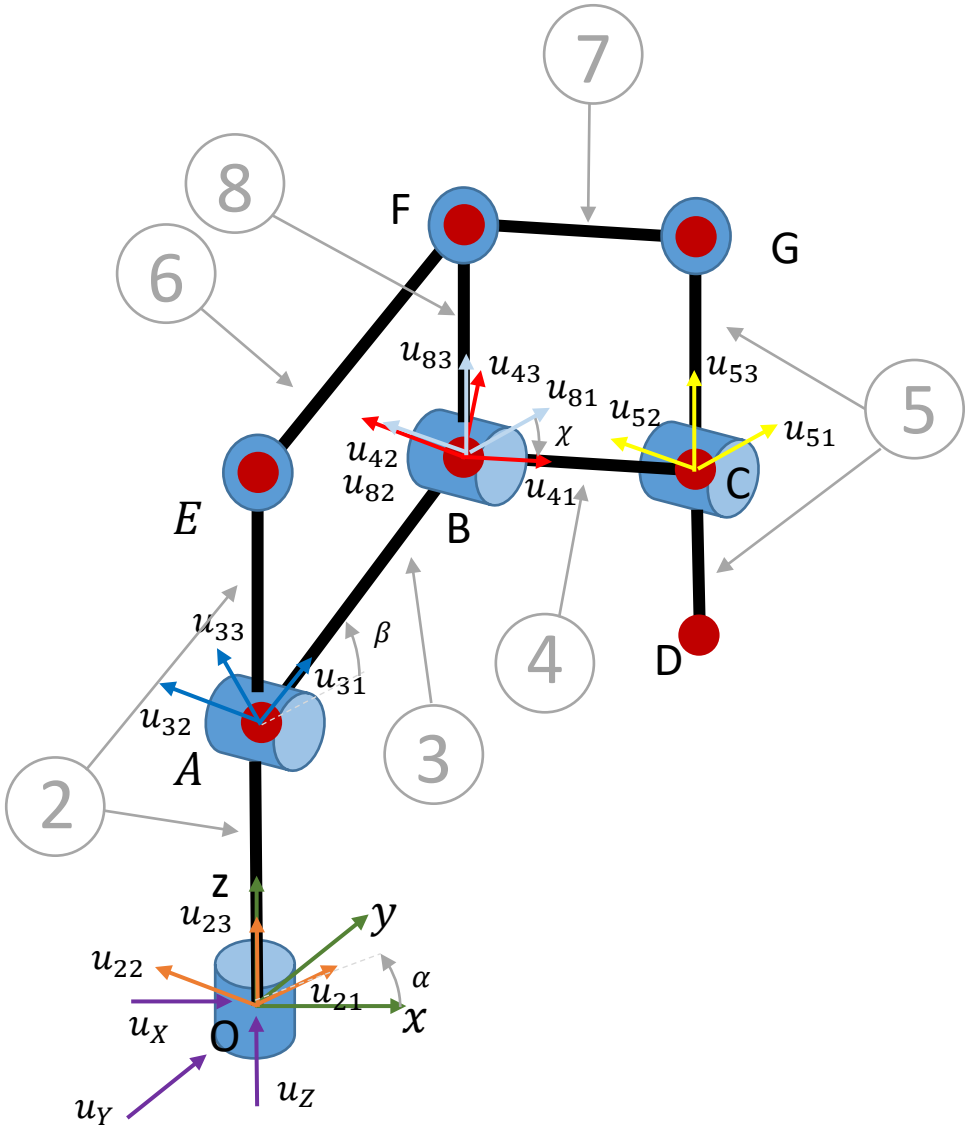
Relación geométrica	Ecuaciones	Número de ecuaciones
Sistema de referencia local de 4	$\lvert\mathbf{u}_{41}\rvert = 1$ $\lvert\mathbf{u}_{42}\rvert = 1$ $\lvert\mathbf{u}_{43}\rvert = 1$ $\mathbf{u}_{41} \cdot \mathbf{u}_{42} = 0$ $\mathbf{u}_{41} \cdot \mathbf{u}_{43} = 0$ $\mathbf{u}_{42} \cdot \mathbf{u}_{43} = 0$	6
Sistema de referencia local de 5	$\lvert\mathbf{u}_{51}\rvert = 1$ $\lvert\mathbf{u}_{52}\rvert = 1$ $\lvert\mathbf{u}_{53}\rvert = 1$ $\mathbf{u}_{51} \cdot \mathbf{u}_{52} = 0$ $\mathbf{u}_{51} \cdot \mathbf{u}_{53} = 0$ $\mathbf{u}_{52} \cdot \mathbf{u}_{53} = 0$	6

Total ecuaciones=24



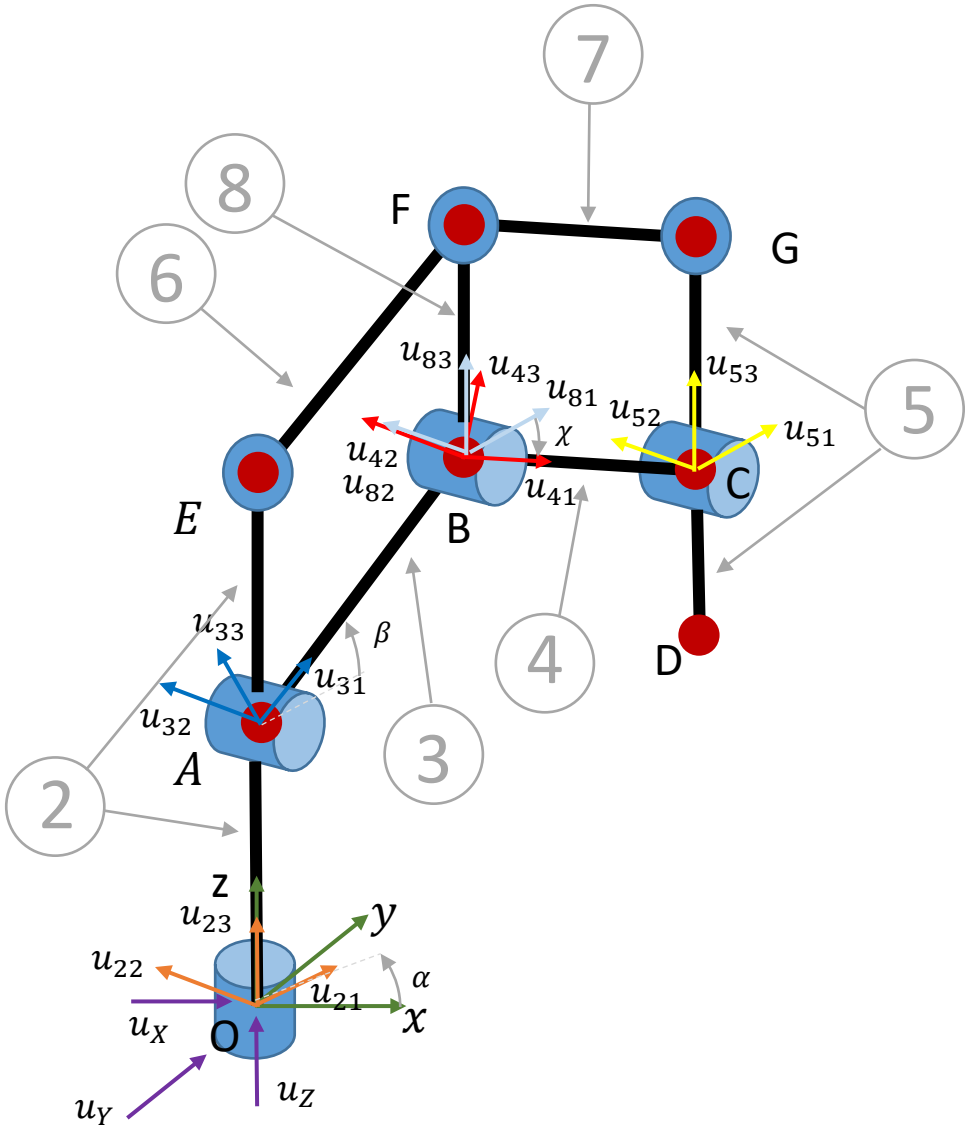
Relación geométrica	Ecuaciones	Número de ecuaciones
Sistema de referencia local de 8	$\lvert\mathbf{u}_{81}\rvert = 1$ $\lvert\mathbf{u}_{82}\rvert = 1$ $\lvert\mathbf{u}_{83}\rvert = 1$ $\mathbf{u}_{81} \cdot \mathbf{u}_{82} = 0$ $\mathbf{u}_{81} \cdot \mathbf{u}_{83} = 0$ $\mathbf{u}_{82} \cdot \mathbf{u}_{83} = 0$	6

Total ecuaciones=30



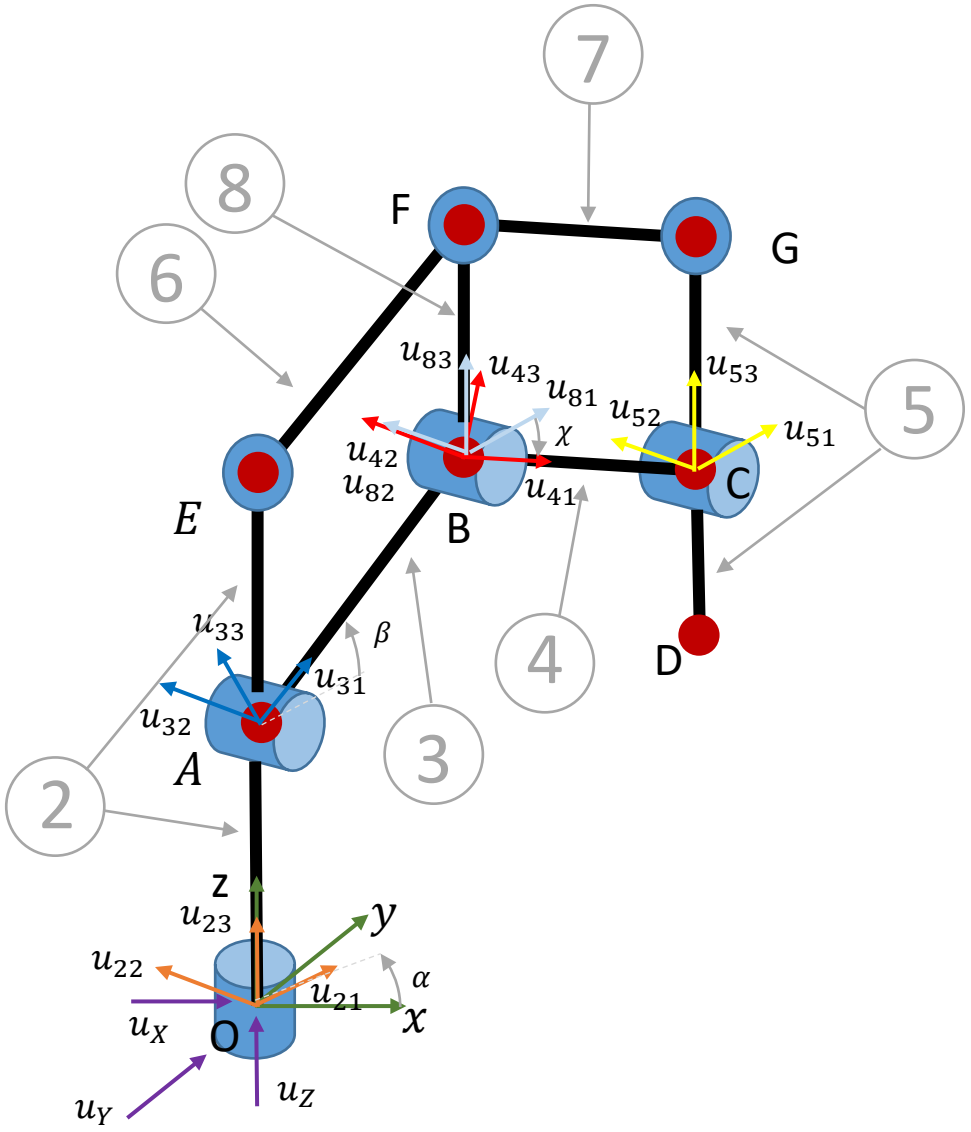
Relación geométrica	Ecuaciones	Número de ecuaciones
A pertenece a 2	$\mathbf{r}_A = \mathbf{r}_O + \mathbf{c}_{A2} \cdot (\mathbf{u}_{21} + \mathbf{u}_{22} + \mathbf{u}_{23})$	3
E pertenece a 2	$\mathbf{r}_E = \mathbf{r}_O + \mathbf{c}_{E2} \cdot (\mathbf{u}_{21} + \mathbf{u}_{22} + \mathbf{u}_{23})$	3
B pertenece a 3	$\mathbf{r}_B = \mathbf{r}_A + \mathbf{c}_{B3} \cdot (\mathbf{u}_{31} + \mathbf{u}_{32} + \mathbf{u}_{33})$	3

Total ecuaciones=39



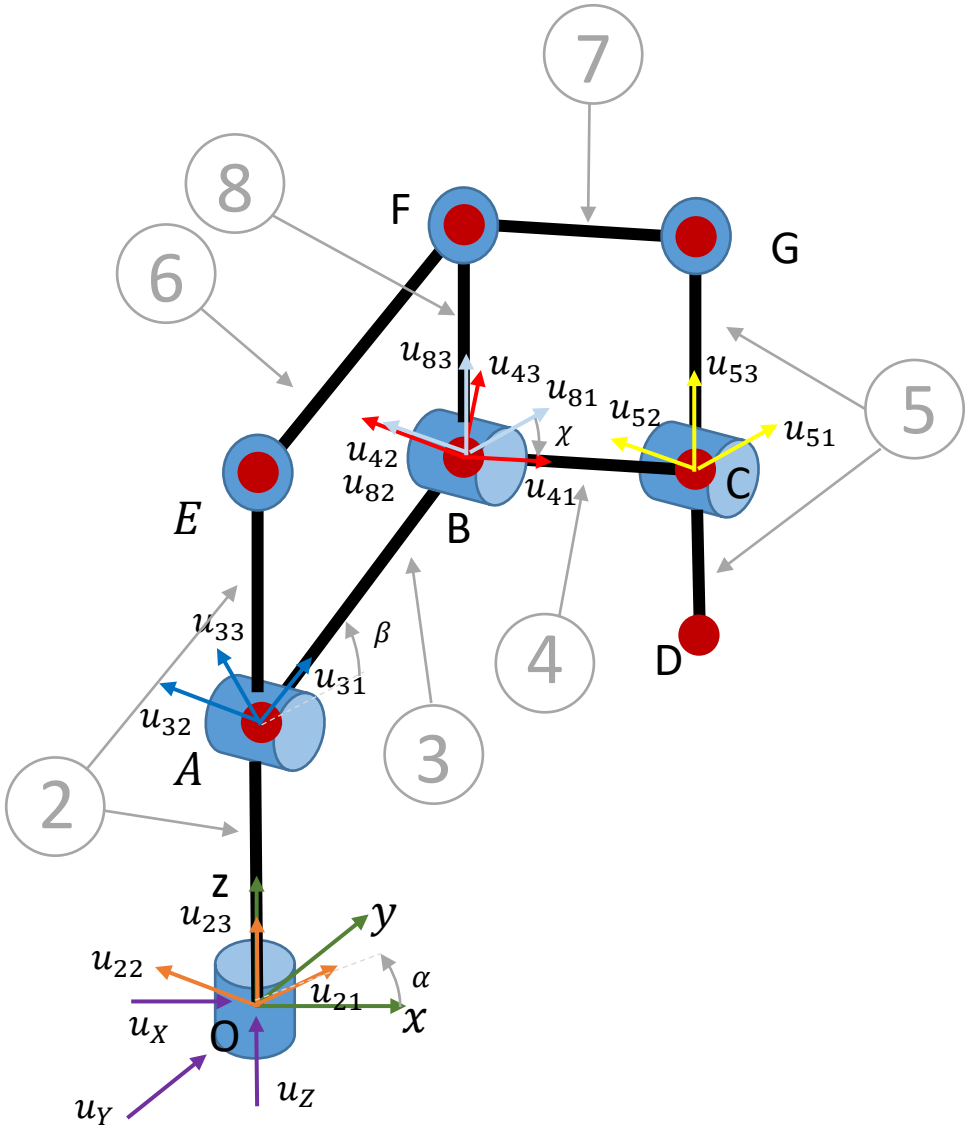
Relación geométrica	Ecuaciones	Número de ecuaciones
C pertenece a 4	$\mathbf{r}_C = \mathbf{r}_B + \mathbf{c}_{C4} \cdot (\mathbf{u}_{41} + \mathbf{u}_{42} + \mathbf{u}_{43})$	3
D pertenece a 5	$\mathbf{r}_D = \mathbf{r}_C + \mathbf{c}_{D5} \cdot (\mathbf{u}_{51} + \mathbf{u}_{52} + \mathbf{u}_{53})$	3
G pertenece a 5	$\mathbf{r}_G = \mathbf{r}_C + \mathbf{c}_{G5} \cdot (\mathbf{u}_{51} + \mathbf{u}_{52} + \mathbf{u}_{53})$	3

Total ecuaciones=48



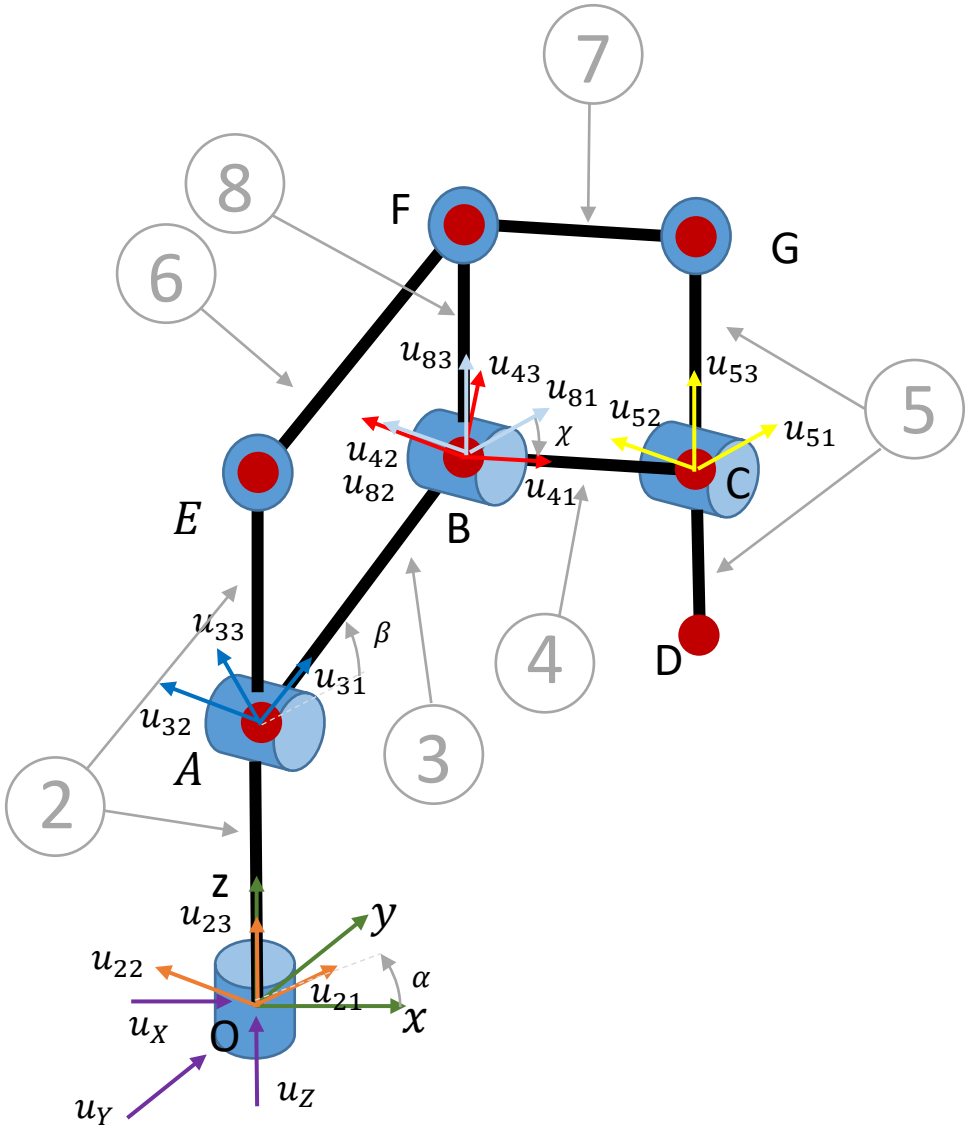
Relación geométrica	Ecuaciones	Número de ecuaciones
F pertenece a 8	$\mathbf{r}_F = \mathbf{r}_B + \mathbf{c}_{F8} \cdot (\mathbf{u}_{81} + \mathbf{u}_{82} + \mathbf{u}_{83})$	3
Distancia constante entre E y F	$\lvert \mathbf{r}_F - \mathbf{r}_E \rvert = L_{EF}$	1
Distancia constante entre F y G	$\lvert \mathbf{r}_G - \mathbf{r}_F \rvert = L_{FG}$	1

Total ecuaciones=53



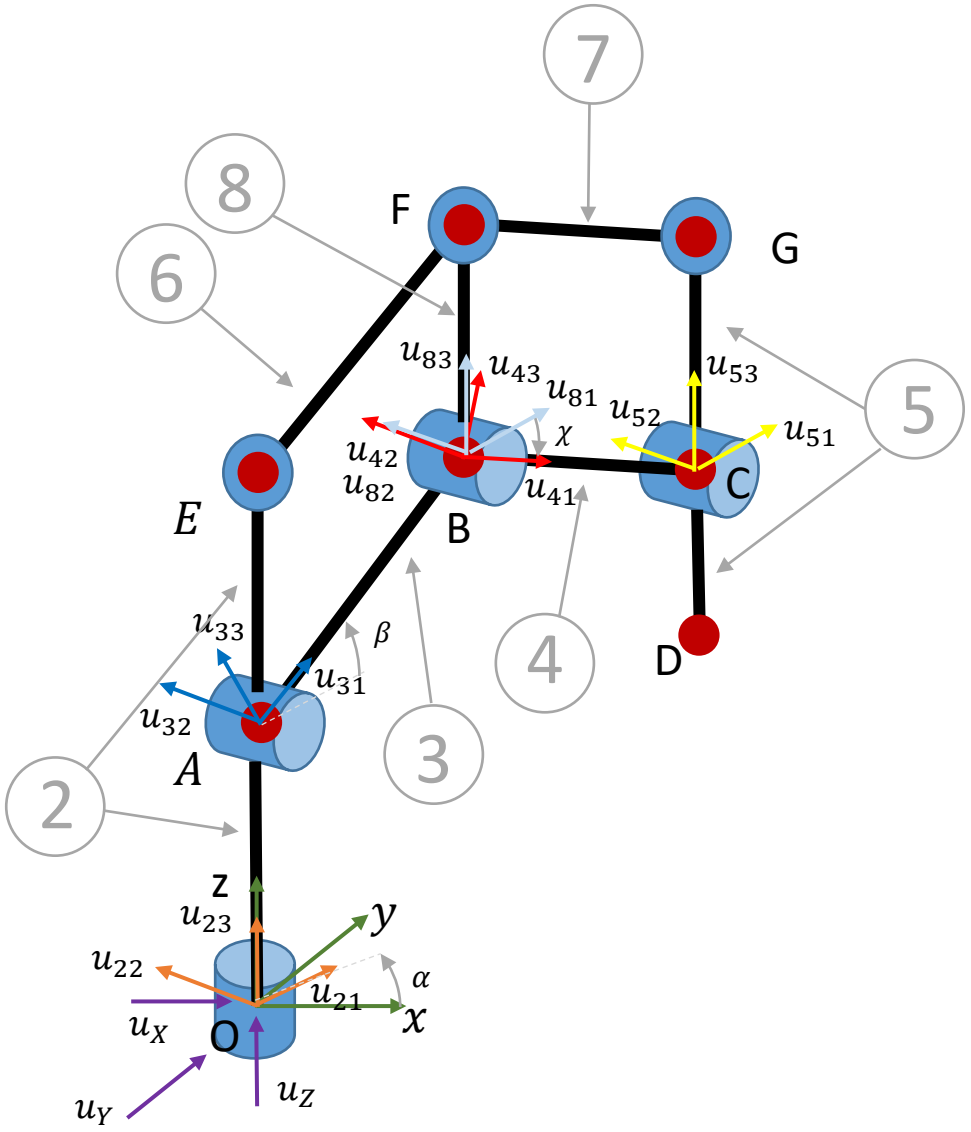
Relación geométrica	Ecuaciones	Número de ecuaciones
Par R en O	$\mathbf{u}_Z \wedge \mathbf{u}_{23} = \mathbf{0}$	2+1
Par R en A	$\mathbf{u}_{22} \wedge \mathbf{u}_{32} = \mathbf{0}$	2+1
Par R en B	$\mathbf{u}_{32} \wedge \mathbf{u}_{42} = \mathbf{0}$	2+1

Total ecuaciones=59+3



Relación geométrica	Ecuaciones	Número de ecuaciones
Par R en C	$\mathbf{u}_{42} \wedge \mathbf{u}_{52} = \mathbf{0}$	2+1
Par R en B	$\mathbf{u}_{42} \wedge \mathbf{u}_{82} = \mathbf{0}$	2+1

Total ecuaciones=63+5



Relación geométrica	Ecuaciones	Número de ecuaciones
Giro absoluto en O	$\mathbf{u}_X \cdot \mathbf{u}_{21} - \cos\alpha = 0$ $\mathbf{u}_Y \cdot \mathbf{u}_{21} - \sin\alpha = 0$	1+1
Giro relativo en A	$\mathbf{u}_{21} \cdot \mathbf{u}_{31} - \cos\beta = 0$ $\mathbf{u}_{23} \cdot \mathbf{u}_{31} - \sin\beta = 0$	1+1
Giro relativo en B	$\mathbf{u}_{81} \cdot \mathbf{u}_{41} - \cos\chi = 0$ $\mathbf{u}_{83} \cdot \mathbf{u}_{41} - \sin\chi = 0$	1+1

Total ecuaciones=66+8